

Kadec–Pełczyński splitting for JBW*-triple preduals

An exact later-literature resolution

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Status. `literature_already_answered`. This note records an explicit later-literature solution; it does not claim a new proof.

The source conjecture

Fernández-Polo, Peralta, and Ramírez, arXiv:1211.4990, Conjecture 4.4 (source-PDF page 24), ask the following. Given a bounded sequence (ϕ_n) in the predual W_* of a JBW*-triple W , do there exist a subsequence $(\phi_{\tau(n)})$ and mutually orthogonal tripotents (e_n) such that

$$\{\phi_{\tau(n)} - \phi_{\tau(n)}P_2(e_n) : n \in \mathbb{N}\} \quad \text{is relatively weakly compact?}$$

Equivalently, the desired decomposition is $\phi_{\tau(n)} = \xi_n + \psi_n$ with $\psi_n = \phi_{\tau(n)}P_2(e_n)$ and $\xi_n = \phi_{\tau(n)} - \psi_n$ [1].

The exact later answer

Peralta and Pfitzner explicitly state in their introduction that their main result “gives a positive answer” to an earlier question and “a proof to [21, Conjecture 4.4]”; their reference [21] is precisely the source paper. Their Theorem 6.1 (supporting-PDF page 16, journal page 92) proves that a subsequence has a decomposition

$$\phi_{n_k} = \psi_k + \xi_k,$$

where the ψ_k are pairwise orthogonal normal functionals and (ξ_k) converges weakly in W_* [2].

For completeness, this statement yields the source’s exact Peirce formulation. Let $e_k = s(\psi_k)$ be the support tripotent. Then the e_k are mutually orthogonal and $\psi_k = \psi_k P_2(e_k)$. The set $K = \{\xi_k : k \in \mathbb{N}\}$ is relatively weakly compact. Proposition 4.3(v) of the supporting paper says that for every orthogonal sequence (e_k) ,

$$\sup_{\xi \in K} \|\xi P_2(e_k)\| \longrightarrow 0.$$

Consequently

$$\phi_{n_k} - \phi_{n_k}P_2(e_k) = \xi_k - \xi_k P_2(e_k).$$

This is a weakly convergent sequence perturbed by a norm-null sequence, hence its range is relatively weakly compact. Thus Conjecture 4.4 is affirmative in full generality.

Scope

The identification is explicit, not inferred only from similar terminology, and the source and supporting PDFs are preserved beside this note. The later paper also asks for reasonable conditions guaranteeing such splitting in arbitrary L -embedded spaces. That broader classification problem is not part of Conjecture 4.4 and is not claimed resolved here.

References

- [1] F. J. Fernández-Polo, A. M. Peralta, and M. I. Ramírez, *A Kadec–Pełczyński dichotomy-type theorem for preduals of JBW^* -algebras*, arXiv:1211.4990; Israel J. Math. 208 (2015), 547–568.
- [2] A. M. Peralta and H. Pfitzner, *The Kadec–Pełczyński–Rosenthal subsequence splitting lemma for JBW^* -triple preduals*, Studia Math. 227 (2015), 77–95, [doi:10.4064/sm227-1-5](https://doi.org/10.4064/sm227-1-5).